Engineering Mechanics 1 · Unit_01 · UNIT 1

DRAFT — UE26ME142A · Unit_01 · UNIT 1 · UE26ME142A Unit 1 Class 2.2 AVSummary

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 1 Class 2.2 AVSummary

1 UE26ME142A Unit 1 Class 2.2 AVSummary

Study focus: UE26ME142A_Unit 1_Class 2.2_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000070 · EV0000071

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Dear student, I welcome all of you for engineer mechanics static classes.

Key details

- My name is Dr.
- Sv Venkadesh from Civil Engineering under Department of Science and Humanities, PS University.
- This is class number four.
- In the last class, we had discussed law of triangle of forces, law of paragraph of forces, cross product, then scalars and vectors, then what are the different.
- Types of vectors based on the point of application of the vector.
- And then what is a sign convention that we use to represent the vectors?
- This was discussed in the last class, that is class number three.
- Cause of the angle made by the two vectors that is V1 and V2, that is this angle.
- plus 2 V1 into V2 into cos of the angle between the two vectors V1 and V2 which I am considering this angle as theta.
- Direction of the resultant that is the angle made by the resultant with respect to one of the vectors, say V1, if this is the angle that I am considering, say alpha.
- Then tan alpha will be equal to we will to get that alpha we will make a construction here.
- This becomes a rectangle triangle, whereas this side will be equal to V1 sorry, it is going to be equal to V two sine of angle theta that is this angle which is same as the angle between V one and V two.
- cos of theta okay.
- The direction of the resultant, the sense of the resultant, and of course the point of the resultant will be the same as the point of application of these two forces, that is this point.

- Or we will go for what is called as rectangular components of a force.
- You this diagram you can see.
- What does this diagram indicate?
- We have a force V which is making an angle of theta with respect to the x axis.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 05:00–09:59

No, to get that we will draw a line parallel to Vx, which is.

Key details

- Along x axis passing through the edge of V, that is from here, so an horizontal line.
- Rectangle, and that's why it is called as a rectangular component.
- However, we will get the component of the force V that is Vx and Vy, which is along the x-axis and along y-axis as V.
- Cos of this angle theta, we get Vx.
- These two are mutually perpendicular to each other, and if we have a force V, we can still get the component V x dash and Vy dash as the cause of this angle.
- And V sign of this angle as V y 1.
- We have Vx and Vy and then if we want to get V which is nothing but the result n we go for what is called as a dot product which I will be explaining it later.
- This is the vector representation of a force.
- What is Vx?
- It's the component of the force along x axis.
- Vy is the component of the force along y axis, whereas I and J are the unit vectors along.
- X axis and along Y axis.
- This is for a force which is two dimensional.
- If we have a three dimensional one, then we will also get plus.
- Vx as I have already informed you that Vx is nothing but V cos theta, that is the component of V along X, and Vy is the component of V along y axis, that is V sine theta.
- But if the two axes are mutually perpendicular to each other, we can get the result and by simply doing this one.
- Taken a force V1 and F force V2.
- Whatever that you are going to use, simply V will be equal to V1 plus V2.
- It is a dot product of.
- The two vectors will give us the magnitude of the resultant of these two forces.
- We have a force V1 of 4 units that is the magnitude of the force which makes an angle of 45 degrees with the y-axis.
- The sense of V1 is moving away from the origin, the origin is this, the intersection of the x-axis and the y axis is the origin, the force is moving away from that.

Example or application from the lecture

- Next let us see what this vector addition is.

- Now there is a simple example that I have taken here.

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EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 10:00–14:59

We also have one more force which is also moving away from the origin, which is V2.

Key details

- The magnitude of V2 is three units, but it makes an angle of thirty degrees with the x-axis below the x-axis.
- The construction is we have constructed a parallel where there is a line.
- Which is drawn parallel to V1 passing through the edge of V2.
- And according to the law of parallelogram of forces, the diagonal passing through the intersection of the two forces would give us the magnitude both in direction.
- And in magnitude.
- V1 into V2 into cos of the angle between the two forces that is this, which is simply equal to 45 degrees, that is this one, 45 plus 30 degrees, which would be 75.
- That gives us the magnitude of the resultant, yes.
- Or if I take only this one, then this total angle will be equal to 30 plus alpha, 10 of 30 plus alpha will be equal to V1, that is this V1,.
- Sine of this angle, which is the total angle, that is 45 plus 35, which is 75, divided by V2, which is 3, plus this component, which is V1.
- cos of 75.
- That is one way of solving or getting the.
- Magnitude, direction, the sense of course will be when you merge these two, you will get the sense.
- That is one way of determining the resultant of these two forces.
- The other way of determining the resultant of these two forces is.
- Simply determine the component of V1 along X and Y and determine the component of V2 along X and Y.
- The dot product of these two would give us the magnitude of the resultant.
- V1 in terms of the unit vector i and j will be v1 is equal to 4.
- Directly this angle is thirty, so we can see that this is thirty.
- This is 4 into cos 45 i is the 4 into cos 45 is the component of V1 along X.
- I is the unit vector along X.
- Now that would be positive because the sense of this when I rotate it and merge it along x axis, the sense is going to be along X, so that is positive.
- When I rotate V2 and merge it with the Y the X axis.

Example or application from the lecture

- This angle is going to be 45 because they have given this angle as 45.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 15:00–19:59

Then I'll get the sense to be same as that of x, so it is positive, so it is 3 into cos thirty degrees, my component of v2 along x, whereas i is the unit vector of along x axis.

Key details

- Next to find out along y, when I rotate this and merge it along y, I get it to be acting downwards, so that's why it is negative.
- Next to determine the component of V2 along y, it is V2 into sine of 30 degrees, so I get it to be acting downwards.
- This would be the component of V2 along y and j is the unit vector along the y axis.
- Of the result, and we simply add V1 and V2.
- The dot product of V1 and V2, we will get the magnitude yes.
- Plus four sine forty five minus three sine thirty degrees J would be the magnitude of the resultant of these two forces.
- I hope you can.
- Find out the magnitude by law of parallel of forces as I have explained, and by resolving the two vectors along their components X and Y, and then adding them or finding out the dot product of these two, and then getting the magnitude of the resultant of these two forces.
- However I will be explaining how to get the.
- Resultant of these two forces.
- The point of application of the force is the origin, that is the intersection of X and Y.
- The force is moving away from the origin.
- By 4, this triangle indicates the slope of this vector.
- We can easily determine what would be this angle, or we know that this is a right angle triangle.
- The hypotenuse is gonna be five, three, four, five method.
- You can use that or you can simply determine this angle by taking this right angle triangle, you can find out this angle theta 1 if I take this angle as theta 1.
- That would be equal to tan inverse of 3 by 4.
- You can find out the magnitude of that.
- Next we have V2 force.
- Of 12 units, the origin is still the intersection of X and Y, and again the force is moving away from the origin.
- Cos theta 1, then this is theta 1, I plus 10 sin theta 1 j.
- Next coming to V2, the component of V2 along X would be in the negative X direction.
- Whereas for the y, if I rotate this and merge it, I get the sense of the component of this force and the positive y direction, so it will be plus, so that will be plus 12 cos 60 degrees J.
- Take the dot product of these two, so you will get the magnitude of the result end.
- We'll stop at this point and we will continue in the next class.
- And this is the fourth class.
- I'll come back to this problem formulation and obtaining solution in my next class.

- However, explain how to get the this.
- So we will stop here and we'll continue in the next class.
- And my name is Dr.

Example or application from the lecture

- The inclination of V one is given as three.
- The inclination of this V2 with the negative x axis is given us sixty degrees.
- One is by using law paragraph of forces, the other one by resolving the given forces along their components and then getting the results.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 1_Class 2.2_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.2_AVSummary__80ec6ef9f2cc872e.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 2.2_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.2_AVSummary__80ec6ef9f2cc872e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 2.2_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.2_AVSummary__80ec6ef9f2cc872e.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 2.2_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.2_AVSummary__80ec6ef9f2cc872e.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md have no readable `.slides.md` text; their original page images are included.
4. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md have no readable `.slides.md` text; their original page images are included.
5. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md have no readable `.slides.md` text; their original page images are included.
7. 3 rendered pages in 004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md have no readable `.slides.md` text; their original page images are included.
8. 3 rendered pages in 005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md have no readable `.slides.md` text; their original page images are included.
9. 2 rendered pages in 006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md have no readable `.slides.md` text; their original page images are included.
10. 3 rendered pages in 008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md have no readable `.slides.md` text; their original page images are included.

11. 2 rendered pages in 009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md have no readable `.slides.md` text; their original page images are included.
12. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md have no readable `.slides.md` text; their original page images are included.
14. 2 rendered pages in 011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md have no readable `.slides.md` text; their original page images are included.
16. 4 rendered pages in 012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md have no readable `.slides.md` text; their original page images are included.
17. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
18. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
19. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
20. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
21. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00018	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...
SRC00019	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00020	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...
SRC00021	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...

C Revision and retrieval practice